

Program - 1

Write and assemble a program to add the following data and then use the simulator to examine the CY flag.

Input Data → 92H, 23H, 66H, 87H, F5H

Org 0000h

MOV A, #92H; Load first i/p data in A register.

MOV B, #23H; Load second i/p data in B register.

ADD A, B ; Add the two data

JNC L1 ; check for carry

INC R0; Increment reg R0, If CARRY occurs

L1: MOV B, A

MOV A, #66H; Load third i/p data in A register.

ADD A, B

JNC L2

INC R0

L2: MOV B,A

MOV A,#87H; Load fourth i/p data in A register.

ADD A,B

JNC L3

INC R0

L3: MOV B,A

MOV A,#0F5H; Load fifth i/p data in A register.

ADD A, B

JNC L4

INC R0

L4: SJMP L4

END

Program 2

Write a program to calculate y where $y = x^2 + 2x + 9$.

x is between 0 and 9 and the look-up table for x^2 is located at the address (code space) of 200H.

Register R0 has the x, and at the end of the program R2 should have y.

Use the simulator to change the x value and single-step through the program, examining the registers as you go.

ORG 000H

MOV DPTR, #200H

MOV A, #03H; loading x value, eg.x=3

MOV R1, A; copy x value to R1

MOV R0,A; copy x value to R0

ADD A, R1; Add with carry(adding to get 2x)

MOV R1, A; R1 will have 2x value

MOV A, R0; Load A with x value

MOVC A, @A+DPTR; Mov from code space (ROM) to A register (get X^2 from lookup table into register A)

ADD A,#09H; x^2+9

ADD A, R1; x^2+2x+9

MOV R2, A

HERE: SJMP HERE

ORG 200H

DB 00H,01H,04H,09H,10H,19H,24H,31H,40H,51H

END

ADDRESS(h)	Data(h)
0200h	00
0201h	01
0202h	04
0203h	09
0204h	10
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